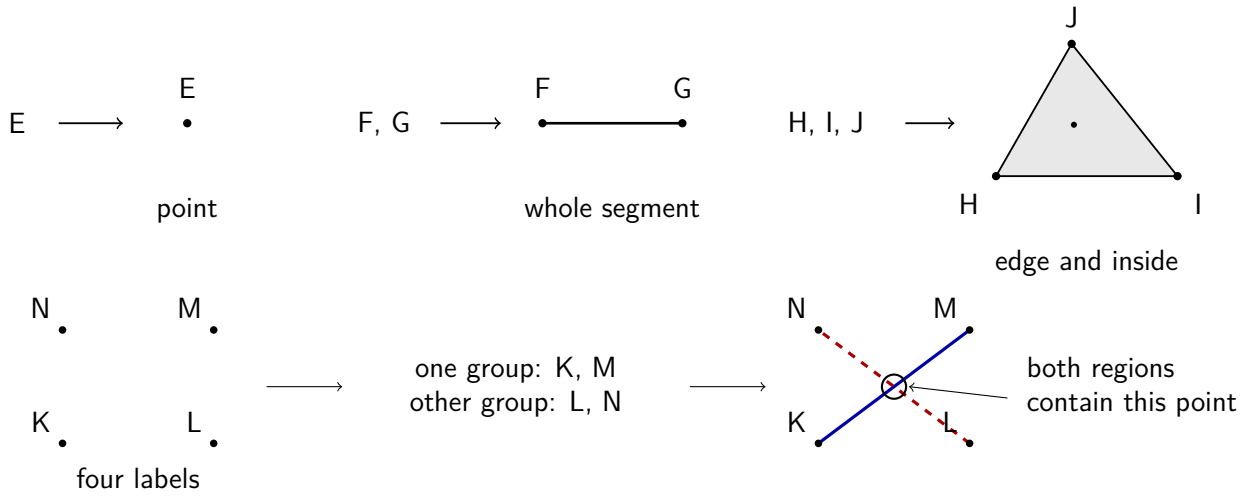
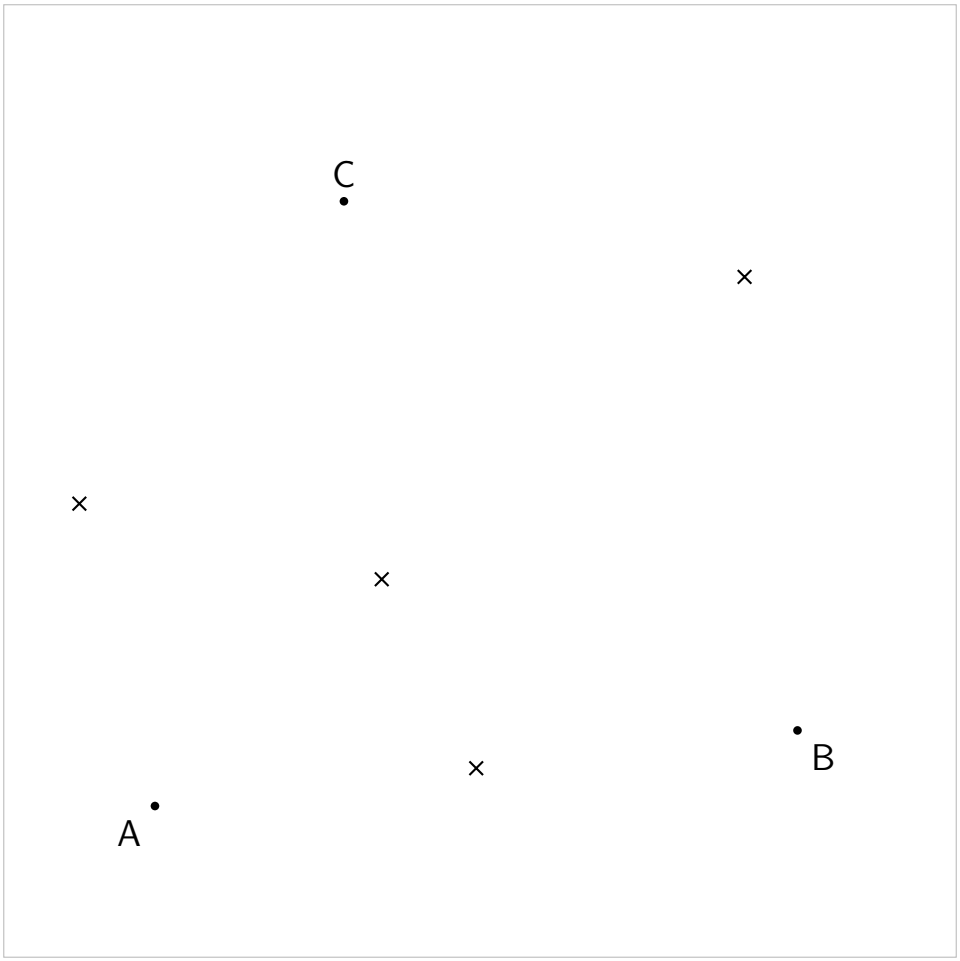
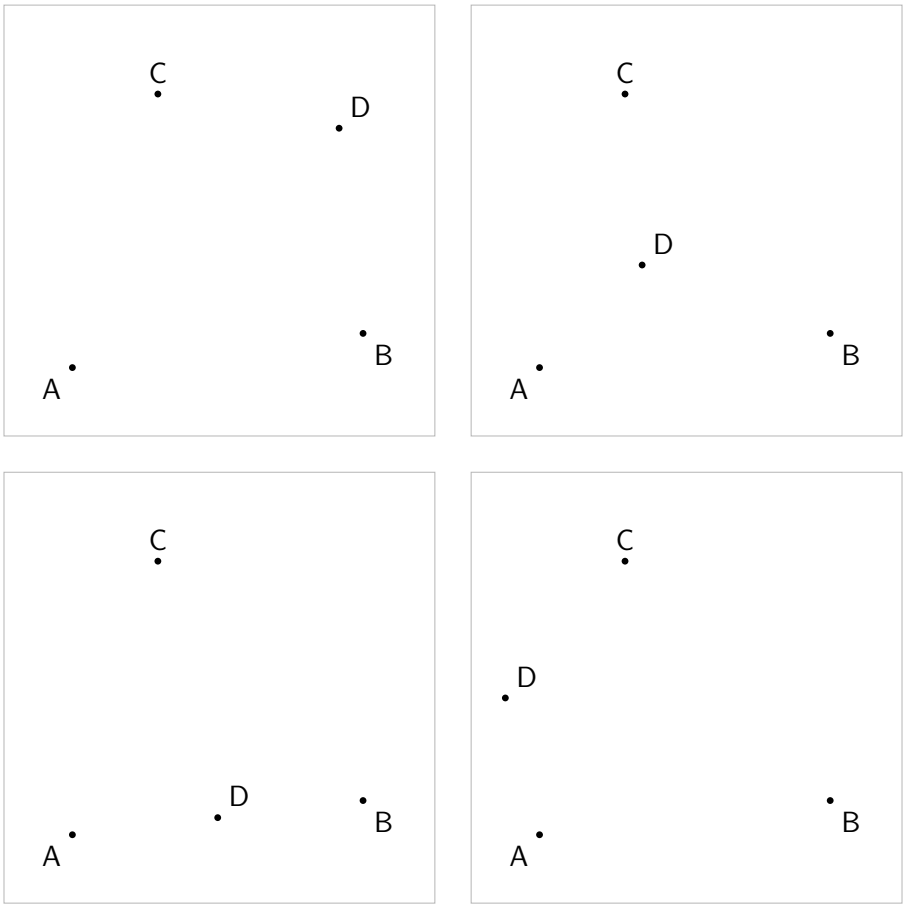
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split.



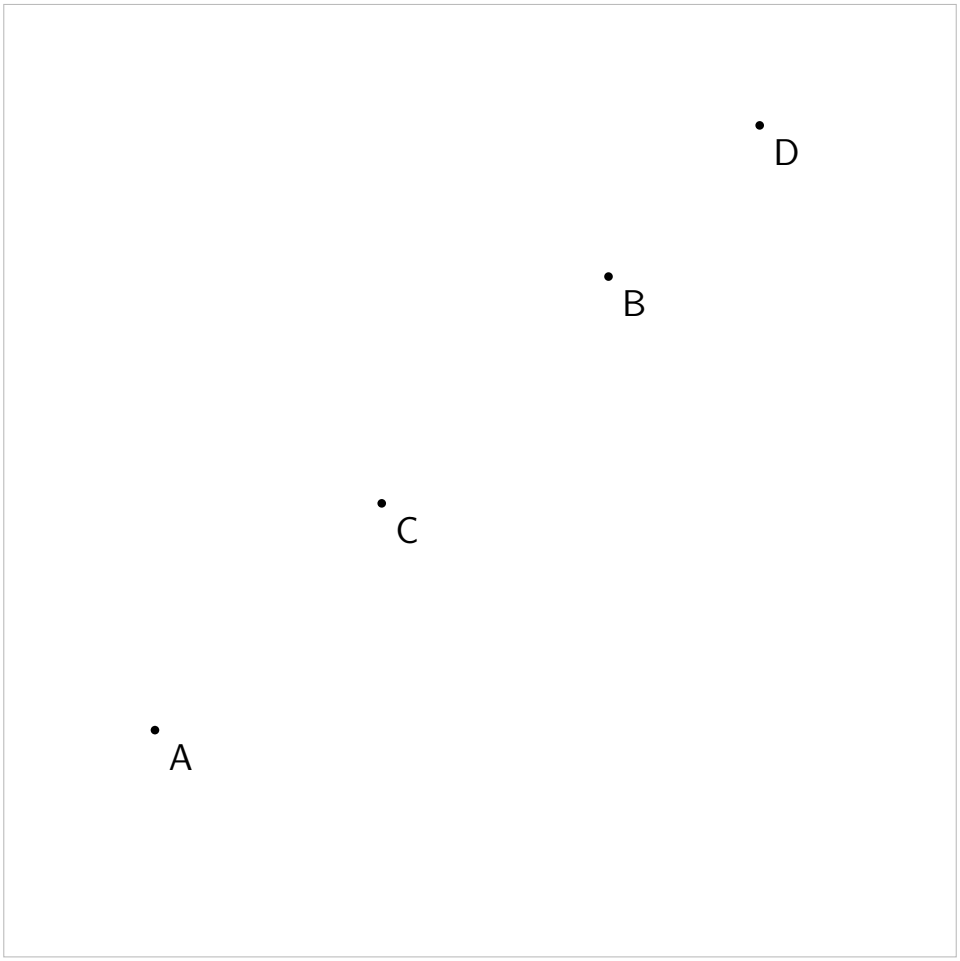
Problem 1: Put D on each cross. For each position, split A, B, C, and D so their two regions meet, and mark a shared point.



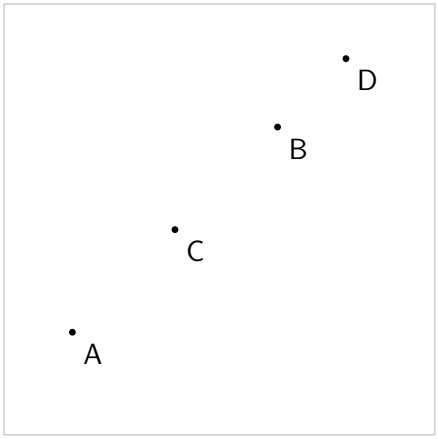
Problem 1: For each position of D, split A, B, C, and D so their two regions meet, and mark a shared point.

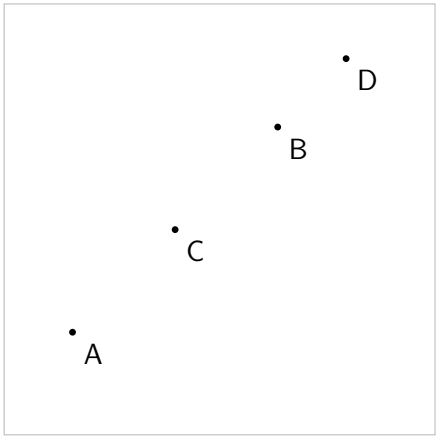


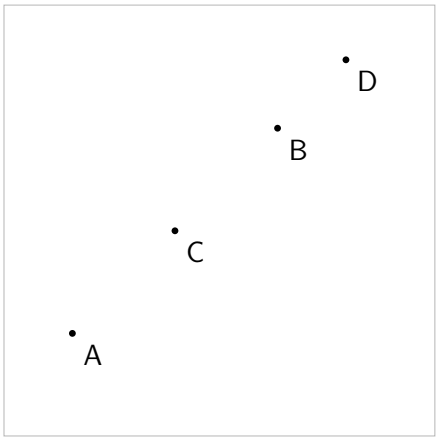
Problem 2: Find every split whose regions meet. Explain how you know your list is complete.

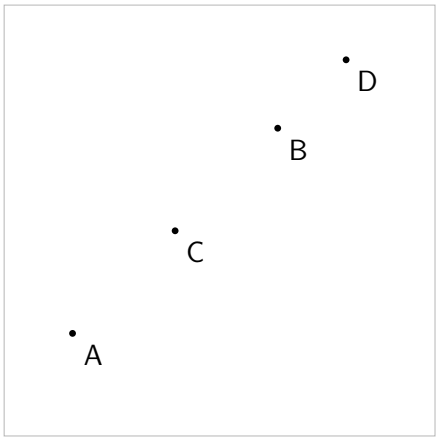


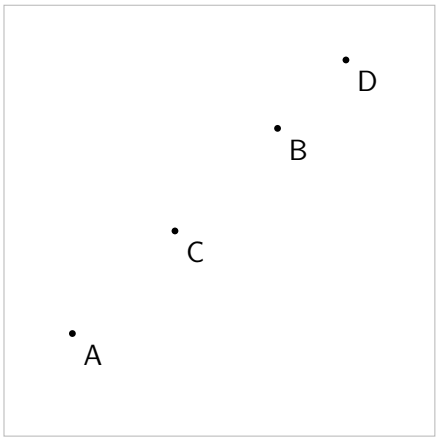
Problem 2: Find every split whose regions meet. Explain how you know your list is complete.

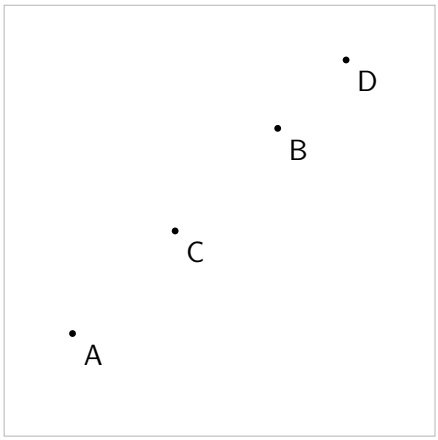












Problem 3: Take turns placing A, B, C, and D in the box. Try to make an arrangement with no successful split. Decide whether your opponent must always be able to find one.